i. Contact exchange and ii. Carbonic acid exchange.

i. **Contact Exchange Theory:**



Figure 11.23: Contact Exchange theory

According to this theory, the ions adsorbed on the surface of root cells and clay particles (or clay micelles) are not held tightly but oscillate within a small volume of space called **oscillation volume**. Due to small space, both ions overlap each other's oscillation volume and exchange takes place (Figure 11.23).

ii. **Carbonic Acid Exchange Theory:**

According to this theory, soil solution plays an important role by acting as a medium for ion exchange. The CO_2 released during respiration of root cells combines with water to form carbonic acid (H_2CO_3). Carbonic acid dissociates into H^+ and HCO_3^- in the soil solution. These H^+ ions exchange with cations adsorbed on clay particles and the cations from micelles get released into soil solution and gets adsorbed on root cells (Figure 11.24).

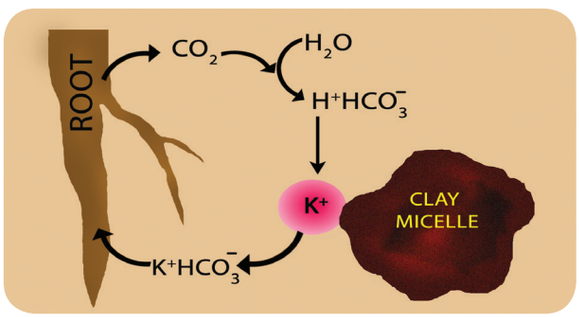


Figure 11.24: Carbonic Acid Exchange theory

11.8.2 Active Absorption

Absorption of ions against the concentration gradient with the expenditure of metabolic energy is called **active absorption**. In plants, the vacuolar sap shows accumulation of anions and cations against the concentration gradient which cannot be explained by theories of passive absorption. Mechanism of active absorption of salts can be explained through carrier concept.

Carrier Concept:

This concept was proposed by **Van den Honert** in 1937. The cell membrane is largely impermeable to free ions. However, the presence of **carrier molecules** in the membrane acts as a vehicle to pick up or bind with ions to form **carrier-ion-complex,** which moves across the membrane. On the inner surface of the membrane, this complex breaks apart releasing ions into cell while carrier goes back to the outer surface to pick up fresh ions (Figure 11.25).

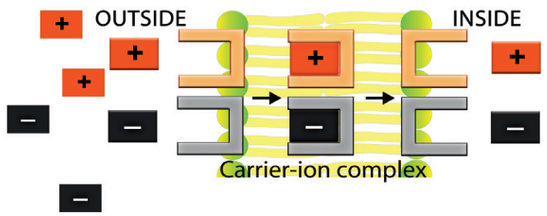


Figure 11.25: Carrier Concept

The concept can be explained using two theories:

1. Lundegardh's Cytochrome Pump Theory: **Lundegardh** and **Burstrom** (1933) observed a correlation between respiration and anion absorption. When a plant is transferred from water to a salt solution the rate of respiration increases which is called as **anion respiration** or **salt respiration**. Based on this observation **Lundegardh** (1950 and 1954) proposed cytochrome pump theory which is based on the following assumptions:

i. The mechanism of anion and cation absorption are different.

ii. Anions are absorbed through cytochrome chain by an active process, cations are absorbed passively.

iii. An oxygen gradient responsible for oxidation at the outer surface of the membrane and reduction at the inner surface. According to this theory, the enzyme *dehydrogenase* on inner surface is responsible for the formation of protons (H^+) and electrons (e^-). As electrons pass outward through electron transport chain there is a corresponding inward passage of anions. Anions are picked up by oxidized cytochrome

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